

Conformal Field Theory

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Preface

This is my notes on Conformal Field Theory.

Chapter 1

Conformal Group

1.1 Conformal Transformation

一般来说在一个流形 M 上定义的场只需要拥有 $Iso(M)$ 的对称性。但现在我们考虑 M 上某些对称性更高的场， M 上所有共形变换诱导出的场变换都是这些场的对称变换。

Remark 1.1.1. 共形变换一般来说不是 M 的等距变换，而是 M 上的共形场的对称变换。

下面先定义一个黎曼/伪黎曼流形 M 上的共形变换，并求出所有共形变换构成的变换群。再将共形群作为场的对称群，从而构造出流形 M 上的经典共形场和量子共形场。

Definition 1.1.2. Let $(M, g), (M', g')$ be smooth manifolds, $\varphi : M \rightarrow M'$ is a diffeomorphism. φ is a conformal transformation between M and M' if $\exists$ a positive function $\Omega : M \rightarrow \mathbb{R}$, s.t.

$$\varphi^* g' = \Omega g \quad (1.1)$$

Choosing definite coordinates, (1.1) could be represented as

$$g'_{\mu\nu}(\varphi(x)) \frac{\partial \varphi^\mu}{\partial x^\rho} \frac{\partial \varphi^\nu}{\partial x^\sigma} = \Omega(x) g_{\mu\nu}(x) \quad (1.2)$$

In physics, we focus on the case in which $M = M'$. If $M = \mathbb{R}^{p,q}$, then (1.2) could be write as

$$\eta_{\mu\nu} \frac{\partial \varphi^\mu}{\partial x^\rho} \frac{\partial \varphi^\nu}{\partial x^\sigma} = \Omega(x) \eta_{\mu\nu} \quad (1.3)$$

(1.3) is the equation that conformal transformation should satisfy.

Conformal transformation is transformation on M , so we could introduce corresponding Killing vectors. We call it conformal Killing vector field.

Definition 1.1.3. $\{\varphi_t : M \rightarrow M | t \in I\}$ is called a conformal flow on M , if φ_t is a conformal transformation on M , $\forall t \in I$.

Definition 1.1.4. Vector Field X on M is called conformal Killing vector field, if the corresponding flow of X is a conformal flow.

Theorem 1.1.5. X is a conformal Killing field on M , the corresponding conformal flow is $\{\varphi_t | \forall t \in I\}$ and the scale factor is $\Omega(x)$, then there exists a smooth function $k \in C^\infty(M)$, satisfying

$$\nabla_\mu X_\nu + \nabla_\nu X_\mu = k g_{\mu\nu} \quad (1.4)$$

and

$$k(x) = \frac{d}{dt} \Omega(\varphi_t(x))|_{t=0} \quad (1.5)$$

Remark 1.1.6. For Killing vector field which is used to describe symmetry of manifolds, it satisfy

$$\nabla_\mu X_\nu + \nabla_\nu X_\mu = 0 \quad (1.6)$$

But for conformal Killing vector field, it satisfy

$$\nabla_\mu X_\nu + \nabla_\nu X_\mu = k g_{\mu\nu} \quad (1.7)$$

Remark 1.1.7. Each possible smooth function k correspond to a possible conformal transformation on M .

Theorem 1.1.8. Smooth function $k \in C^\infty(M)$ correspond to a conformal transformation on d -dimensional manifold M , iff k satisfies

$$(d-2)\partial_\mu \partial_\nu k - g_{\mu\nu} \partial^\lambda \partial_\lambda k = 0 \quad (1.8)$$

1.2 Conformal Group of $\mathbb{R}^{p,q}$

1.2.1 Infinitesimal Conformal Transformation

1.2.2 Lie Algebra of Conformal Group

1.2.3 Conformal Group

Theorem 1.2.1. If $\dim(M) = 1$, then $\text{Conf}(M) = \text{Diff}(M)$.

Theorem 1.2.2. If $p+q \geq 3$, then $\text{Conf}(\mathbb{R}^{p,q}) = SO(p+1, q+1)$.

Theorem 1.2.3. If $p+q = 2$, then $\text{Conf}(\mathbb{R}^{p,q}) = SL(2, \mathbb{C})/\mathbb{Z}_2$.

1.3 Quantization

1.3.1 Central Extension of Group

1.3.2 Quantization of Symmetry

1.3.3 Quantization of Conformal group

Chapter 2

Representation of Virasoro Algebra

2.1 Witt Algebra

2.2 Virasoro Algebra

2.3 Representations of Virasoro Algebra

Chapter 3

Conformal Field Theory

3.1 Classical Conformal Field Theory

3.2 Quantum Conformal Field Theory

Chapter 4

General Properties of Conformal Field Theory

Chapter 5

$(1+1)$ -dimensional Conformal Field Theory

Chapter 6

Minimal Model

Chapter 7

WZW Model

Chapter 8

CFT on Non-Trivial Manifold

Chapter 9

Supersymmetric Conformal Field Theory

Chapter 10

Boundary Conformal Field Theory

Chapter 11

Conformal Bootstrap

Chapter 12

AdS/CFT Duality